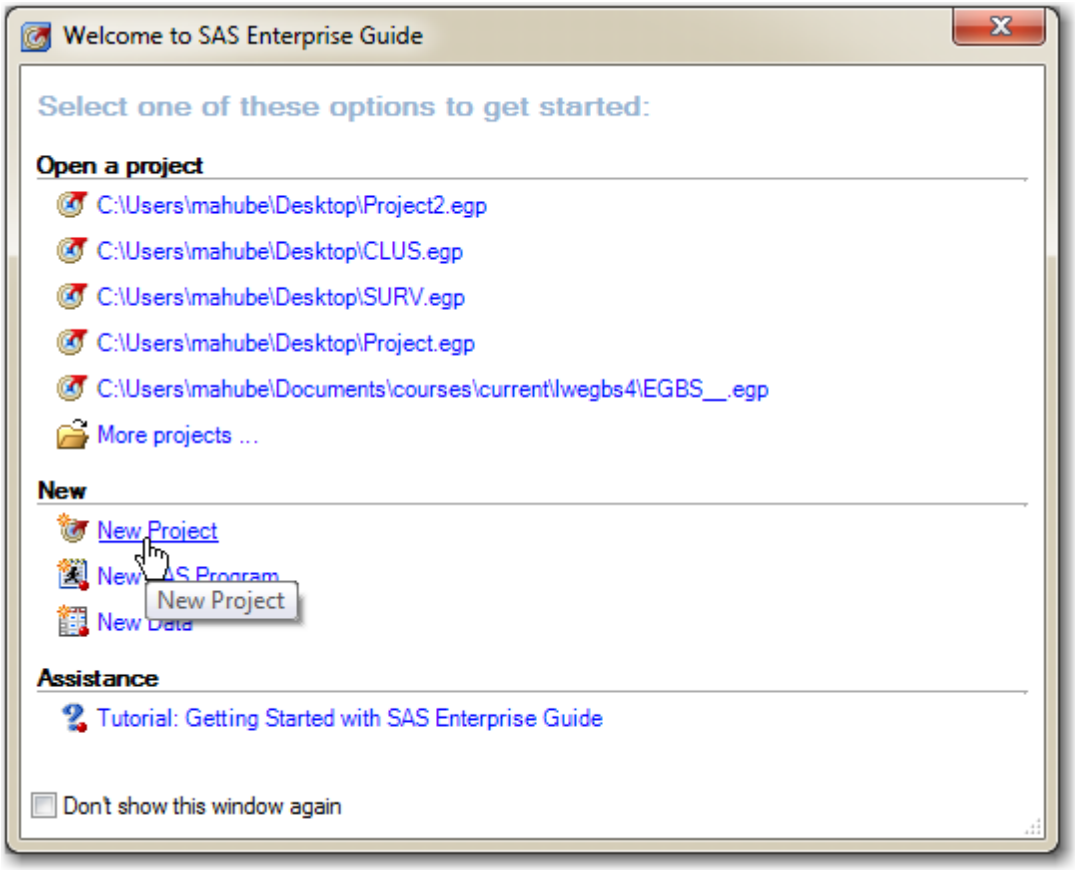
This demo contains:

- **Creating the Demo Datasets**
- **Producing Summary Statistics Tables and Plots**
- **Examining Distributions**
- **Confidence Intervals**
- **Hypothesis Testing**

Creating the Demo Datasets

Create the data sets for the course by running the SAS program in the class folder. Then use the Summary Statistics task to create descriptive statistics.

1. When you open SAS Enterprise Guide, you see a dialog box that gives you several options. Select **New Project**.



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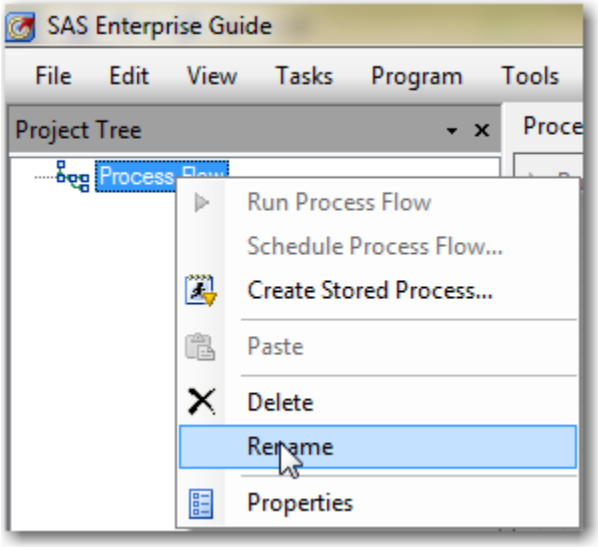
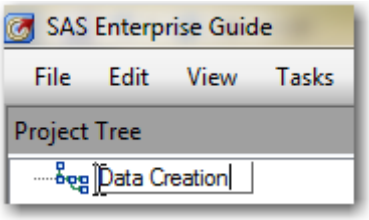
2. Rename the process flow by right-clicking the **Process Flow** icon in the **Project Tree** pane and clicking **Rename** in the drop-down menu.



3. Type **Data Creation** over the words Process Flow. This will enable you to organize one project into separate process flows throughout the course.

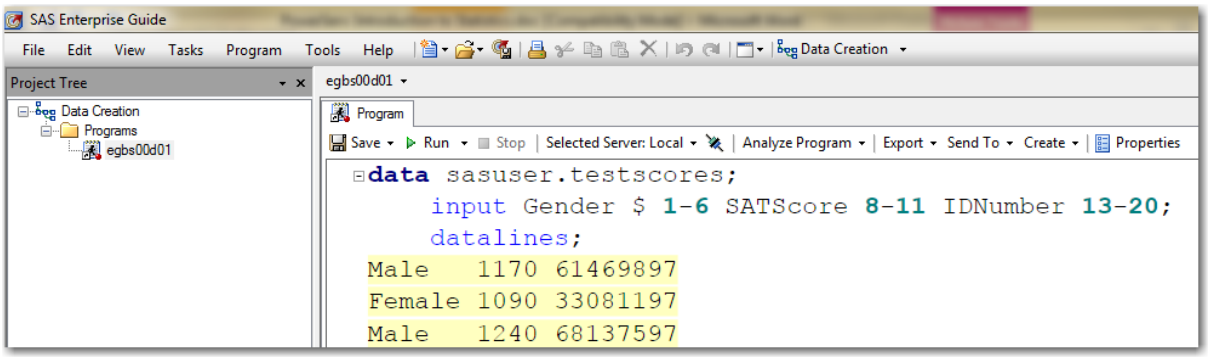


4. Select **File** ⇨ **Open** ⇨ **Program** or the button and then select **Program**.

5. Navigate to the location of the **egbs00d01.sas** file, as given to you by your instructor, and open that file by double-clicking the filename.

   This SAS program generates all of the data sets that are used in demonstrations and exercises for this course.

   Partial file contents:



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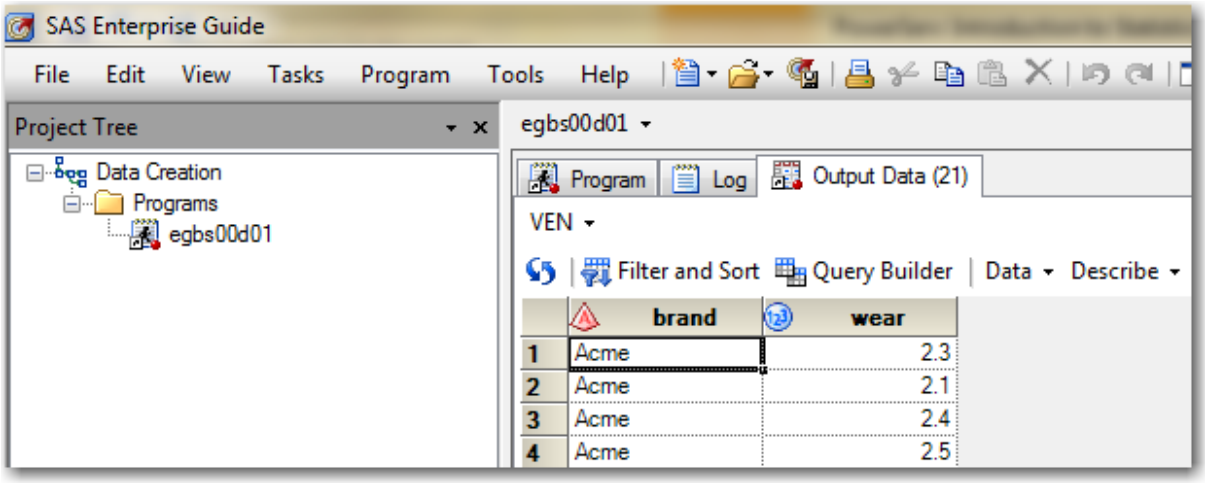
6. Above the program, click **Run**.

SAS Enterprise Guide creates code and submits it to a SAS session called a *server*. The SAS server at your site might have a different name. If you have SAS installed on the same machine as SAS Enterprise Guide, the server is usually named ***local***.

A series of SAS data sets will be created. The **VEN** data set is opened in the right-most pane in the SAS Enterprise Guide work area because it was the last data set created.



7. Close the data table view by clicking the in the upper right corner of the pane.

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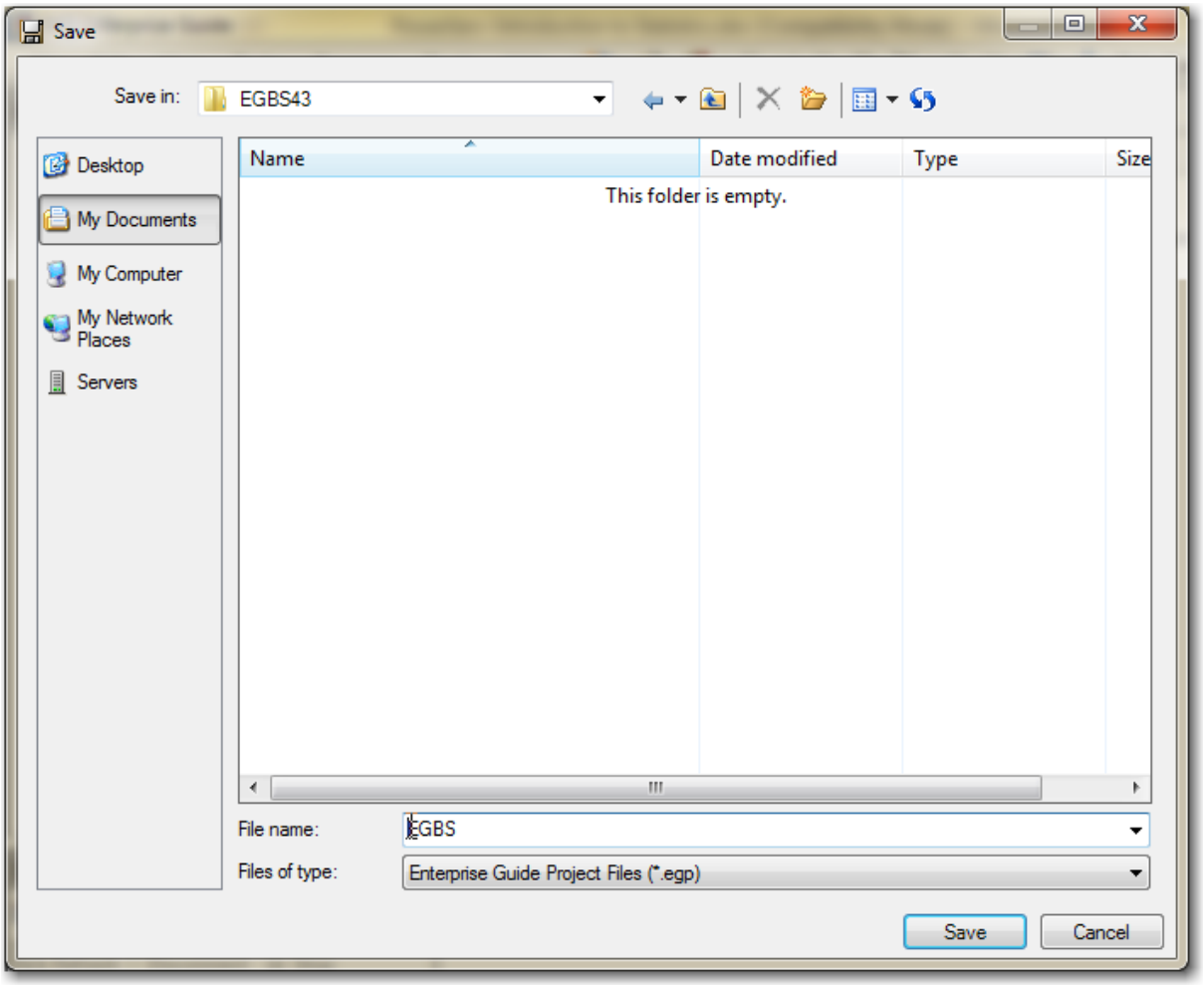
8. Save the project by selecting **File** ⇨ **Save Project As**. Save it in the location specified by your instructor and give it the name **EGBS.egp**. Click **Save**.



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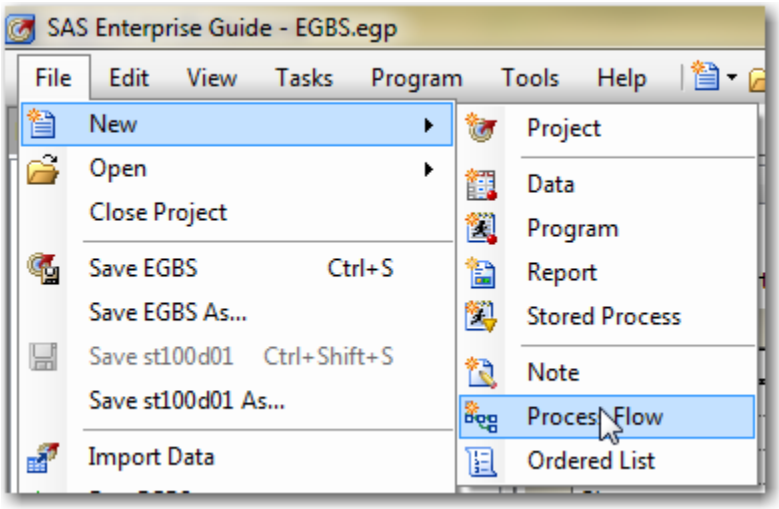
Producing Summary Statistics Tables and Plots

You are ready to begin your analysis. Locate the data set **TESTSCORES** in the Process Flow or Project Explorer and examine its contents. Use the Summary Statistics task to generate descriptive statistics for **SATScore**.

1. Select **File** ⇨ **New** ⇨ **Process Flow** to create a new process flow in the same project.

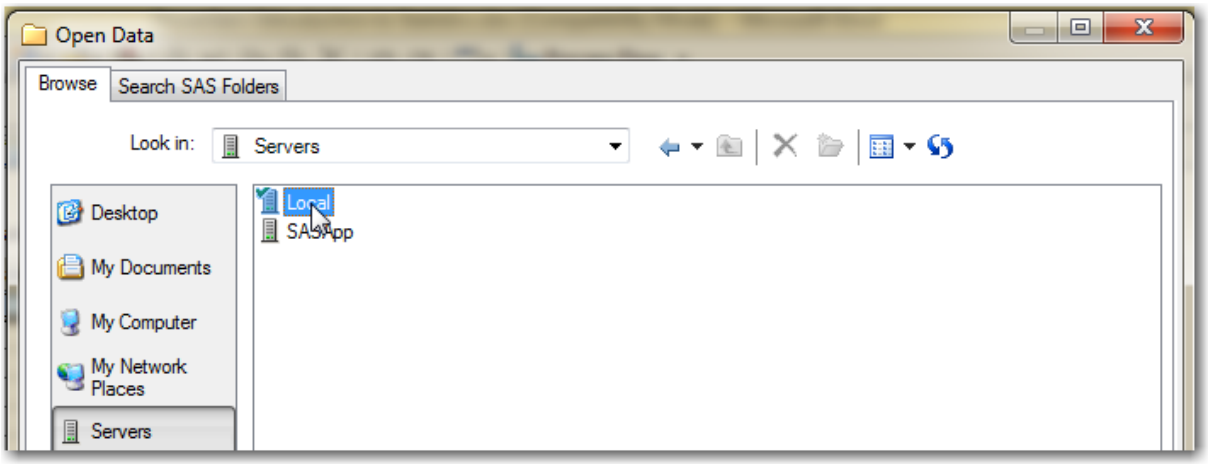


2. Rename this process flow **Introduction to Statistics**.

3. Select **File** ⇨ **Open** ⇨ **Data** or, alternatively, click and then select **Data**.

4. Select **Servers** on the left.

   a. Double-click **Local**.



   b. Double-click **Libraries**.

   c. Navigate to the SASUSER library.

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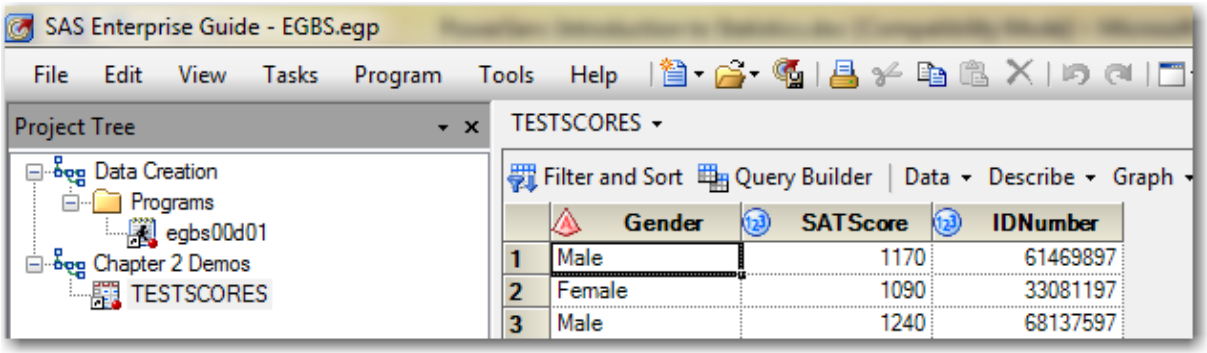
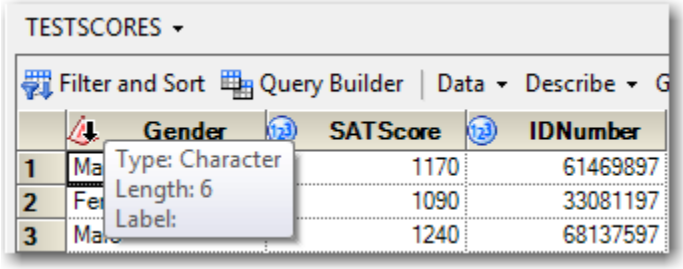
5. Open **TESTSCORES** by double-clicking it or by highlighting it and clicking **Open**.

   The data table opens automatically. You can close it after you look at it.

   Partial Listing



   There are three variables in the **TESTSCORES** data set. One variable, **Gender**, is a character variable that contains the gender of the student. The other two variables, **SATScore** and **IDNumber**, are numeric variables that contain the SAT combined verbal and quantitative score and an identifying code for each student.



6. Above the data table, select **Describe** ⇨ **Summary Statistics** from the drop-down menus.

   If you close the data table first, then you will have to select **Tasks** ⇨ **Describe** ⇨ **Summary Statistics** from the top menu bar.

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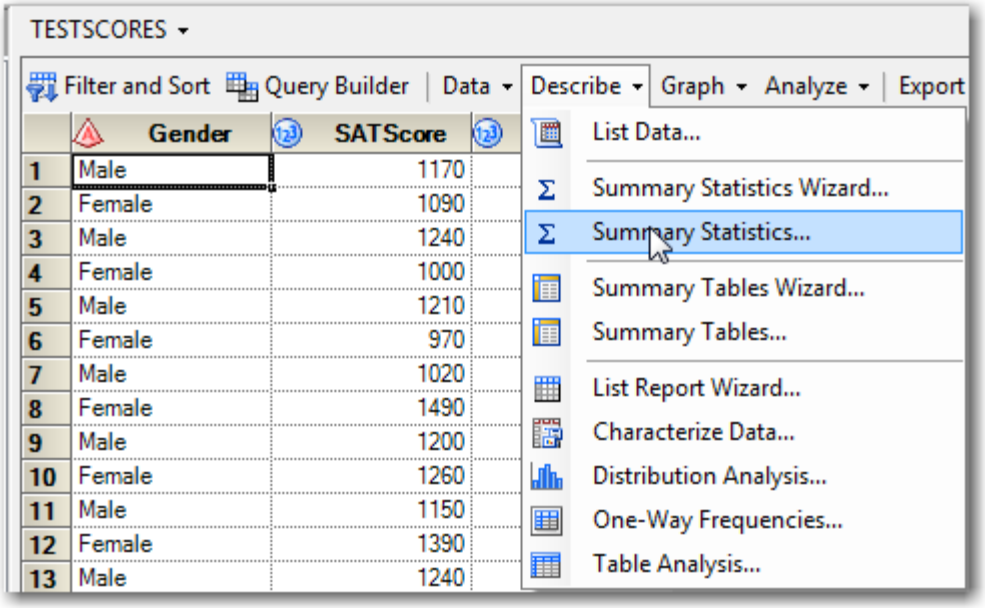
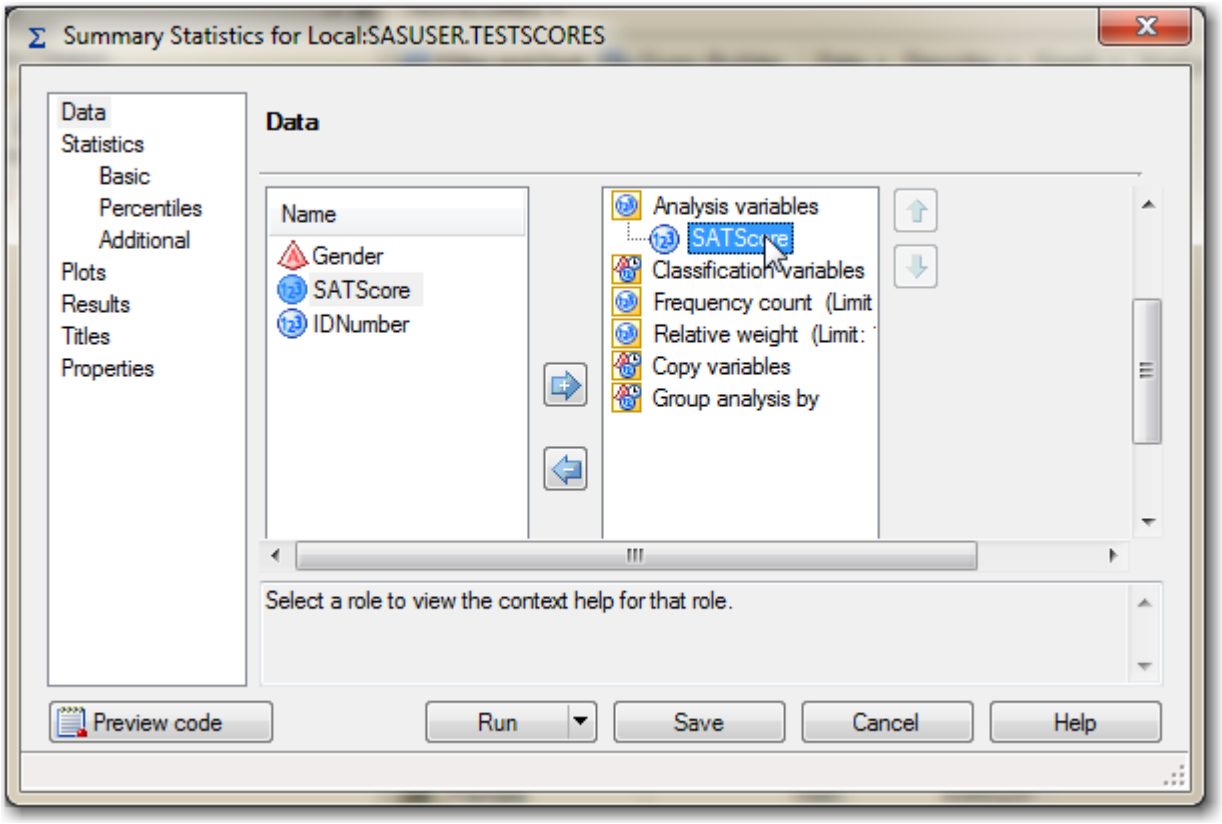


7. With **Data** selected on the left, drag the variable **SATScore** from the **Variables to assign** pane to the analysis variables role in the **Task roles** pane, as shown below:



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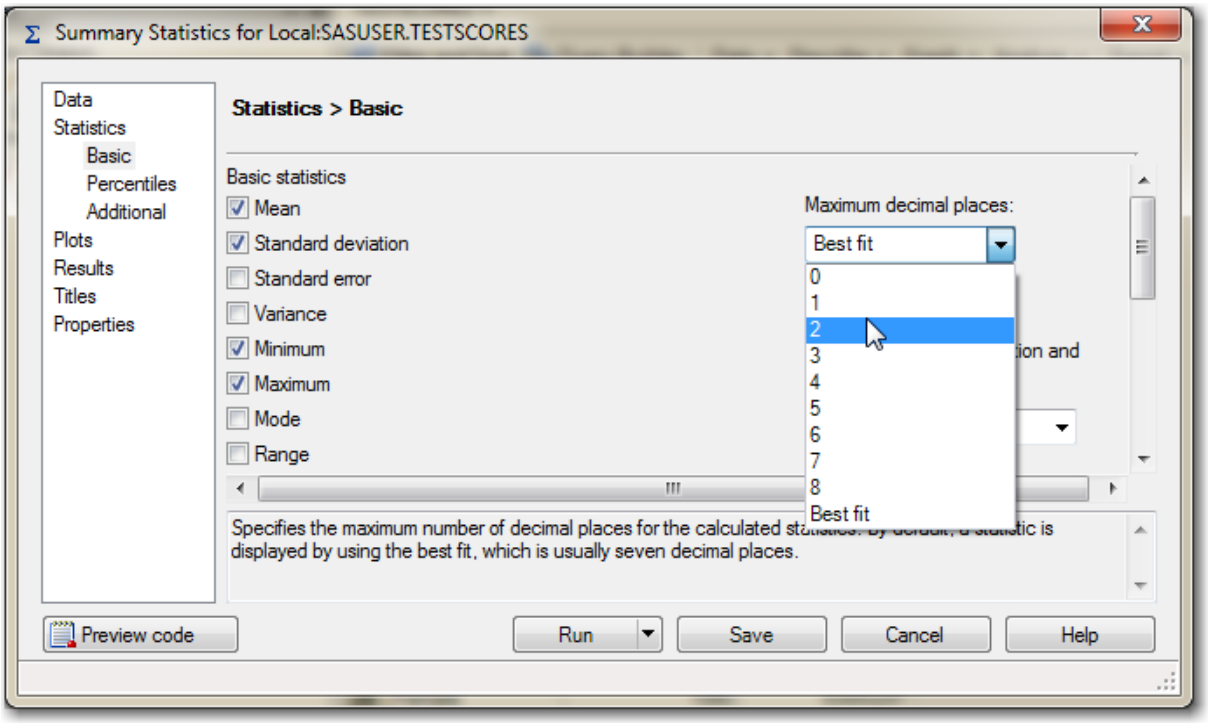
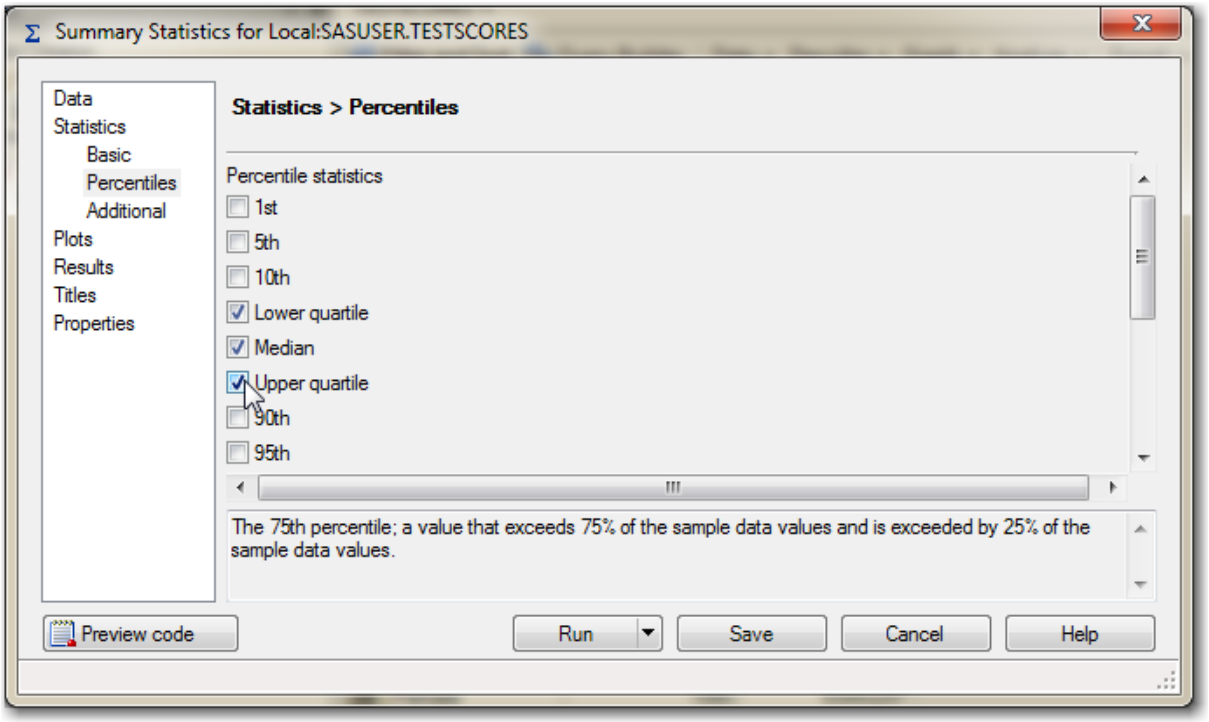
8. Select **Basic** under **Statistics** on the left. Leave the default basic statistics. Change **Maximum decimal places** to **2**.



9. Select **Percentiles** on the left. Under **Percentile statistics**, check the boxes for **Lower quartile**, **Median**, and **Upper quartile**.



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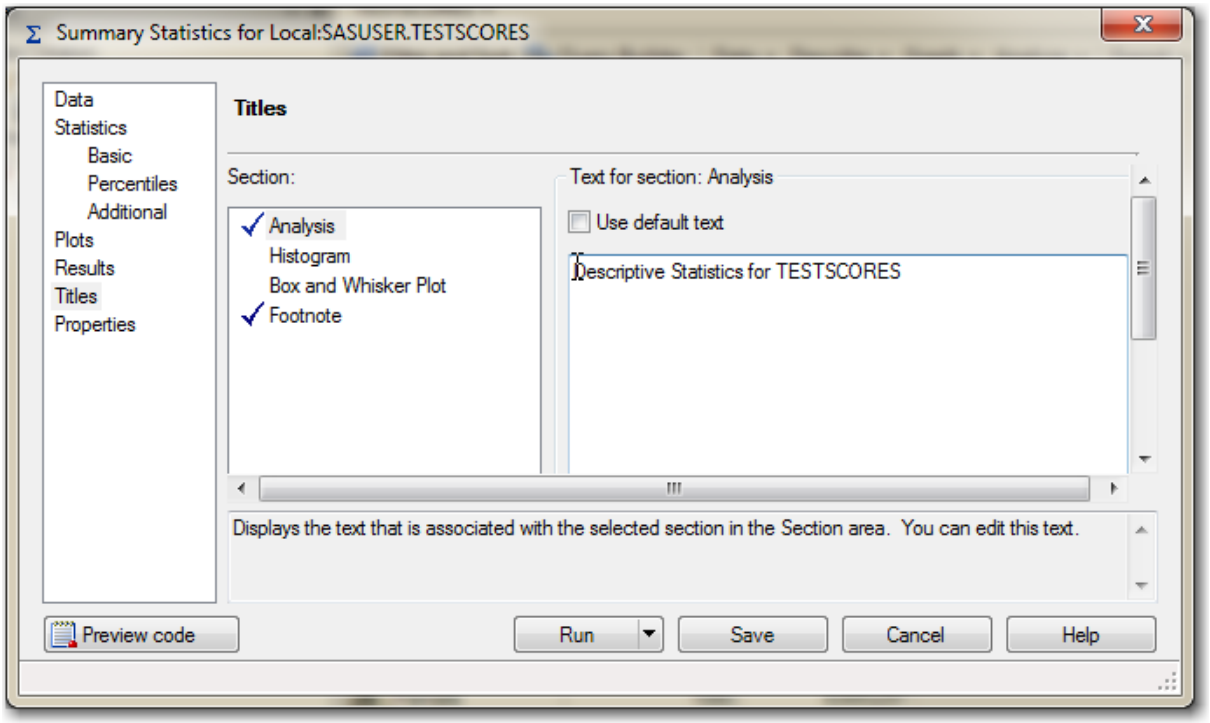
10. Select **Titles** on the left. Deselect **Use default text**. Select the default text in the box and type **Descriptive Statistics for TESTSCORES**. Leave the default footnote text.



11. Click **Run** to run the analysis.

    The report is shown below:

Analysis Variable: SATScore							
Mean	**Std Dev**	**Minimum**	**Maximum**	**N**	**Lower Quartile**	**Median**	**Upper Quartile**
1190.63	147.06	890.00	1600.00	80	1085.00	1170.00	1280.00

    The mean is 1190.63, which is not exactly the 1200 that the school board had set as a goal. The standard deviation is 147.06. The range is 710 (1600 – 890) and the interquartile range is 110 (1280 – 1170).

12. Save the project by selecting **File** ⇨ **Save EGBS** or click .

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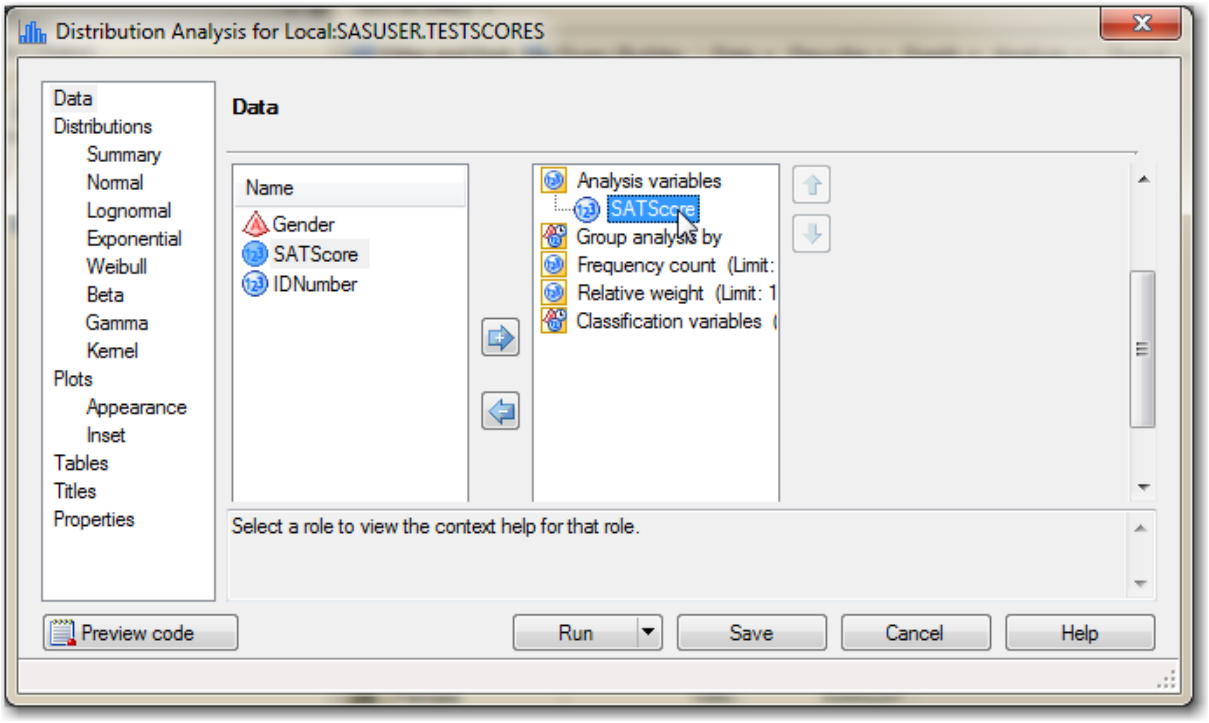
Examining Distributions

This demonstration illustrates how to create statistical tables, histograms, normal probability plots, and box plots using the Distribution Analysis task.

1. Click the **Input Data** tab under **Summary Statistics** from the previous demonstration to show the **TestScores** data table.

2. Select **Describe** ⇨ **Distribution Analysis**.

3. With **Task Roles** selected on the left, drag and drop **SATScore** to the analysis variables role.



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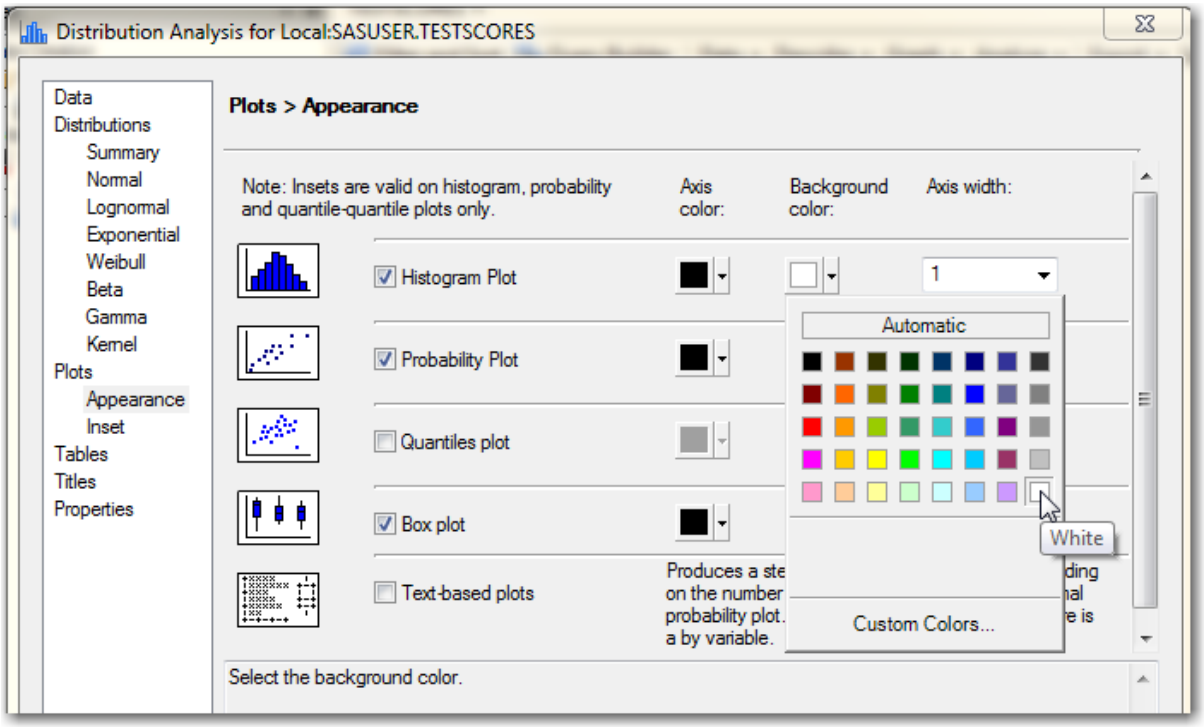
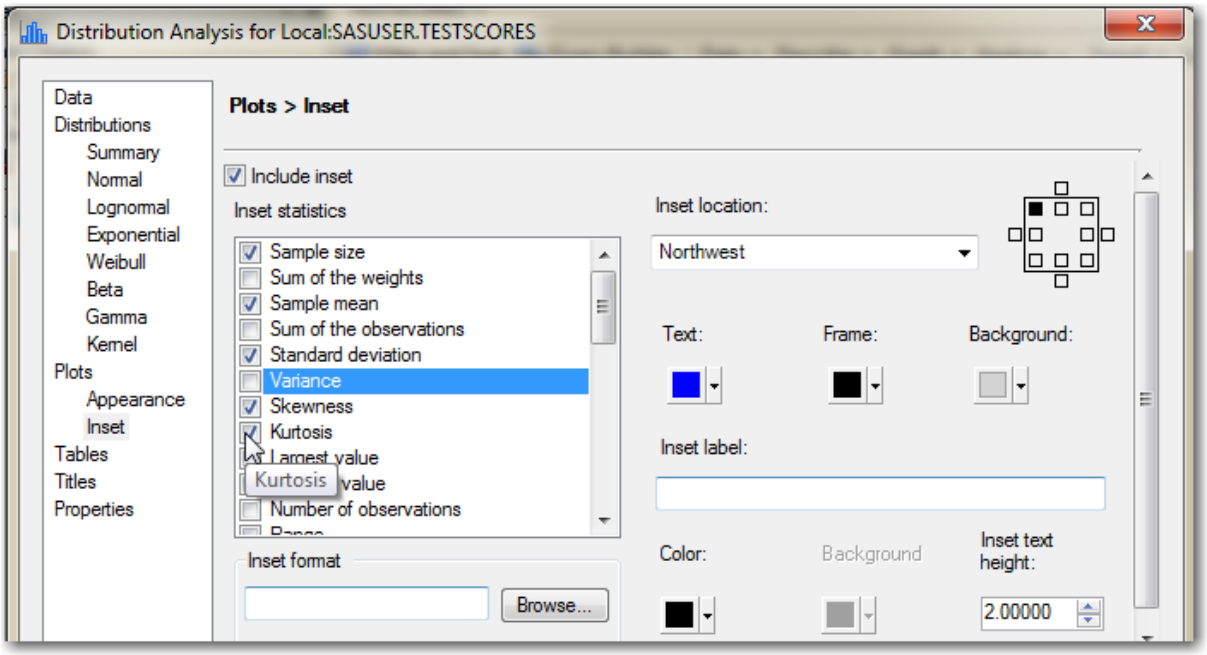
4. Select **Appearance** under **Plots** on the left, and check the box next to **Histogram Plot**, **Probability Plot**, and **Box Plot**. Change the background color in each case to white.



5. Select **Inset** under **Plots** on the left and check the boxes for **Sample Size**, **Sample Mean**, **Standard Deviation**, **Skewness**, and **Kurtosis**.



6. Click **Browse** to change the format for the inset statistics.

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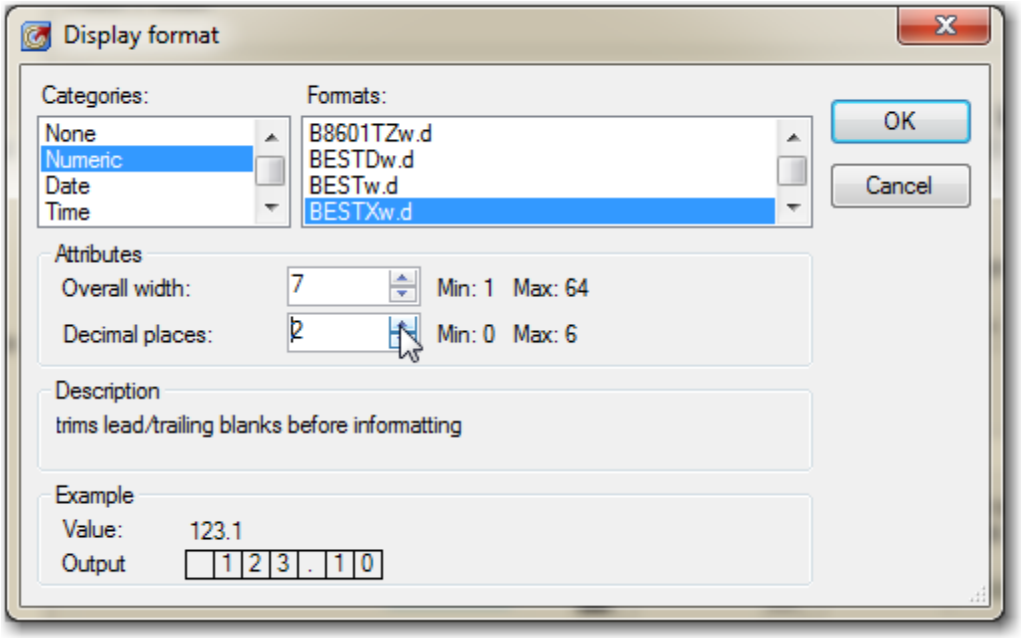
7. Select **Numeric** under **Categories**. Find the BESTXw.d format and assign an overall width of **7** with **2** decimal places. This process will limit the reported output in the inset to seven columns total, with two assigned after the decimal point.



8. Click **OK**.

9. In order to draw a diagonal reference line for the normal probability plot and a normal curve for the histogram, select **Normal** under **Distribution** on the left. Then check the box for **Normal** and change the color to dark red and the width to **3**.

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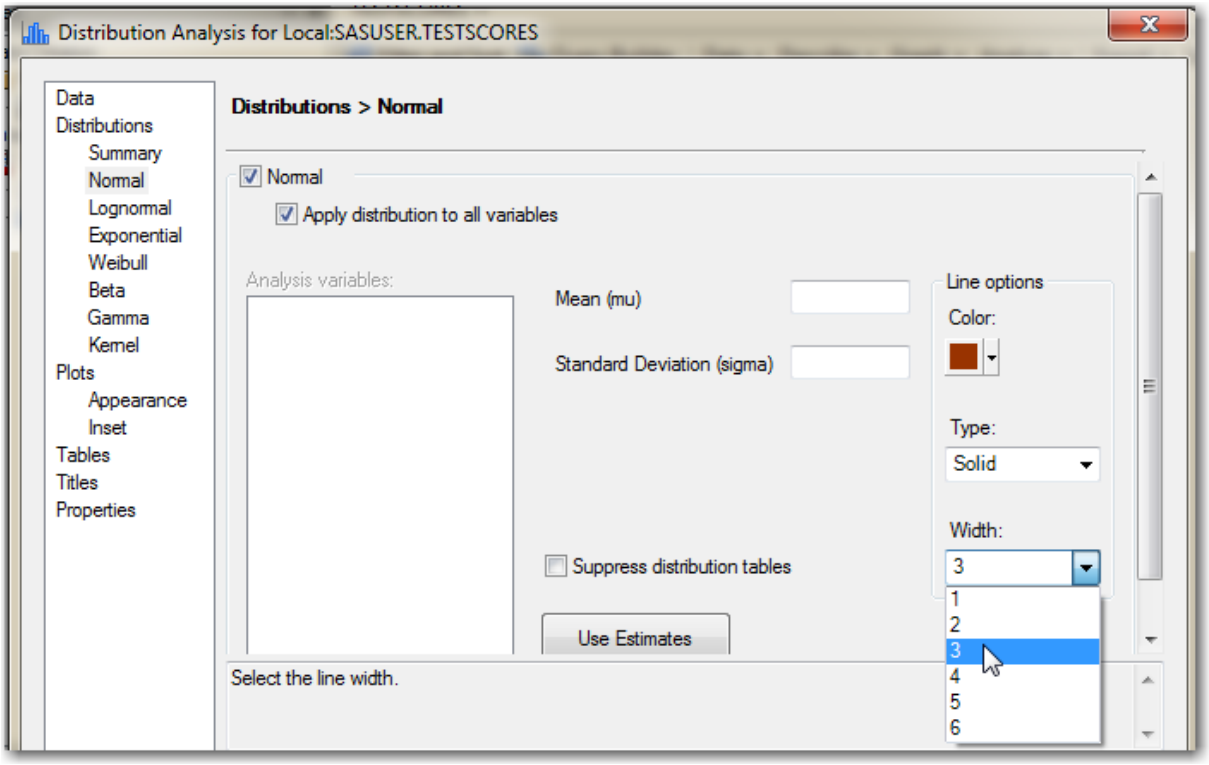
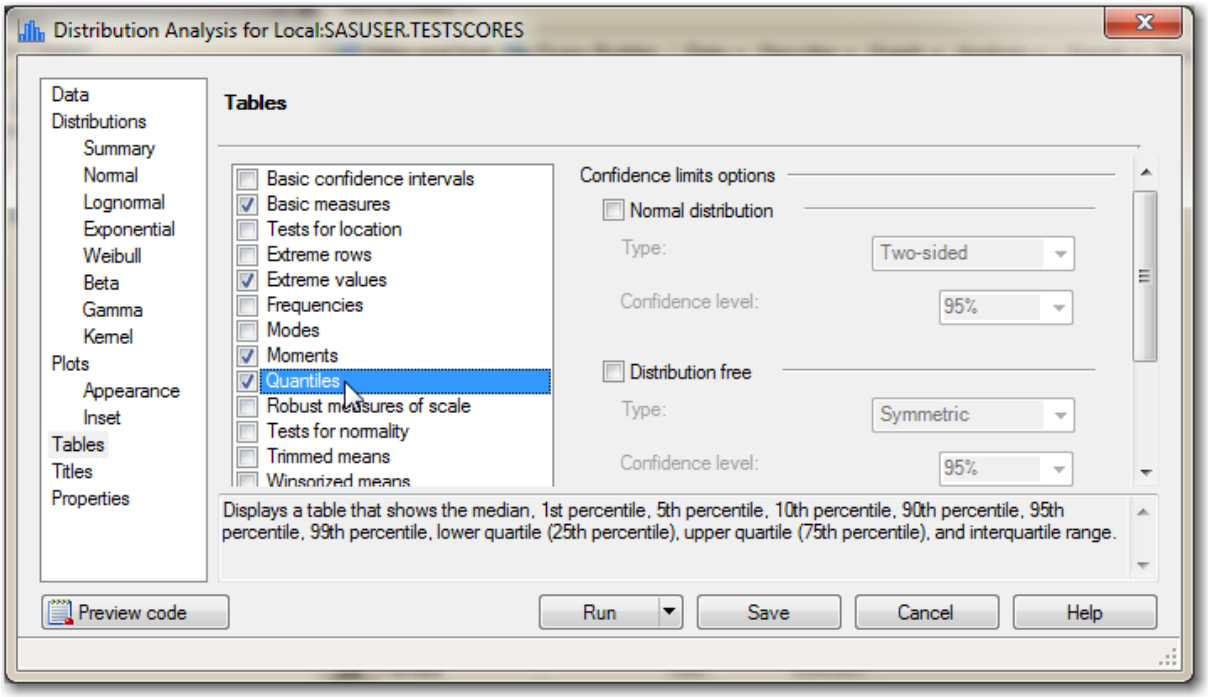


10. Select **Tables** on the left. Deselect the boxes for **Basic confidence intervals** and **Tests for location**. Select the boxes for **Extreme values**, **Moments** and **Quantiles**. (You might need to click twice to select and deselect the tables.)



11. Change the titles and footnotes if desired.

12. Click **Run**.

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Moments			
N	80	**Sum Weights**	80
Mean	1190.625	**Sum Observations**	95250
Std Deviation	147.058447	**Variance**	21626.1867
Skewness	0.64202018	**Kurtosis**	0.42409987
Uncorrected SS	115115500	**Corrected SS**	1708468.75
Coeff Variation	12.3513656	**Std Error Mean**	16.4416342

Basic Statistical Measures			
Location		**Variability**	
Mean	1190.625	**Std Deviation**	147.05845
Median	1170.000	**Variance**	21626
Mode	1050.000	**Range**	710.00000
		Interquartile Range	195.00000

Quantiles (Definition 5)	
Quantile	**Estimate**
100% Max	1600
99%	1600
95%	1505
90%	1375
75% Q3	1280
50% Median	1170
25% Q1	1085
10%	1020
5%	995
1%	890
0% Min	890

Extreme Observations			
Lowest		**Highest**	
Value	**Obs**	**Value**	**Obs**
890	69	1490	8
910	74	1520	42
970	6	1520	54
990	51	1590	70
1000	4	1600	25

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Extreme Values					
Lowest			Highest		
Order	Value	Freq	Order	Value	Freq
1	890	1	39	1390	1
2	910	1	40	1490	1
3	970	1	41	1520	2
4	990	1	42	1590	1
5	1000	1	43	1600	1

The tabular output indicates the following:

- the mean of the data is 1190.625. This is approximately equal to the median (1170), which indicates the distribution is fairly symmetric.
- the standard deviation is 147.058447, which means that the average variability around the mean is approximately 147 points.
- the distribution is slightly skewed to the right (Skewness = +0.64).
- the distribution has slightly heavier tails than the normal distribution (Kurtosis = +0.42).
- the student with the lowest score is observation (row number) 69, with a score of 890. The student with the highest score is row number 25, with a score of 1600 (highest possible score for the SAT.

In the Quantiles table, Definition 5 indicates that the Distributions task is using the default definition for calculating percentile values. You can modify the SAS code to specify one of five methods. These methods are listed in an appendix.

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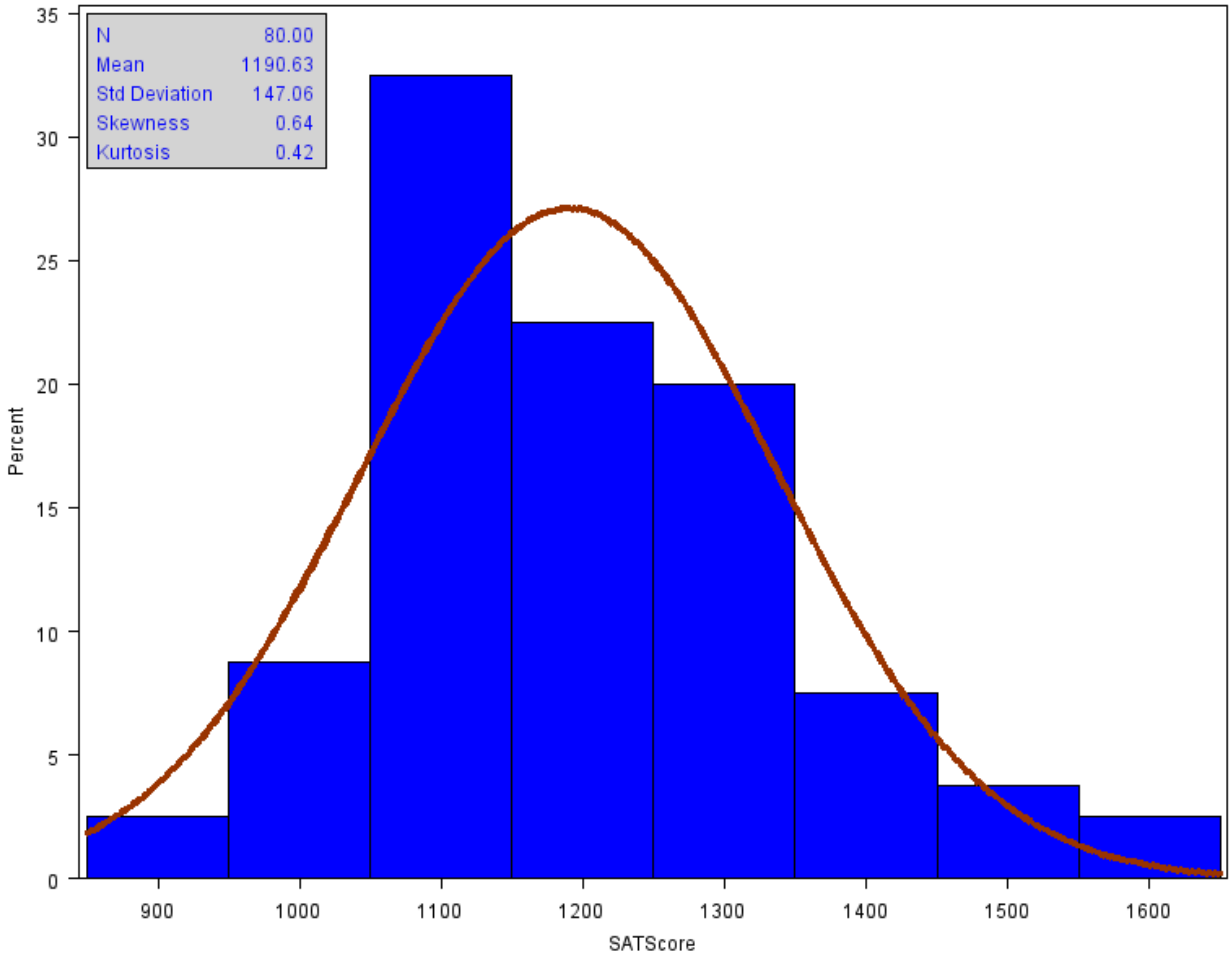

The bin identified with the midpoint of 1100 has approximately 33% of the values. The skewness and kurtosis values are reported in the inset. The kernel density curve is a smoothed version of the histogram and can be used to compare the approximate sample distribution to a normal distribution. In this case, the distribution of the observed data seems to approach normality.

Asking for the normal reference curve for the histogram also produces a set of tables relating to assessing whether the distribution is normal or not. There is a table with three tests presented: the Kolmogorov-Smirnov; Anderson-Darling; and Cramer-von Mises. In each case, the null hypothesis is that the distribution is normal. Therefore, high *p*-values are desirable.

Parameters for Normal Distribution		
Parameter	**Symbol**	**Estimate**
Mean	Mu	1190.625
Std Dev	Sigma	147.0584

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Goodness-of-Fit Tests for Normal Distribution				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.08382224	**Pr > D**	>0.150
Cramer-von Mises	**W-Sq**	0.09964577	**Pr > W-Sq**	0.114
Anderson-Darling	**A-Sq**	0.70124822	**Pr > A-Sq**	0.068

Quantiles for Normal Distribution		
	Quantile	
Percent	**Observed**	**Estimated**
1.0	890.000	848.516
5.0	995.000	948.735
10.0	1020.000	1002.162
25.0	1085.000	1091.436
50.0	1170.000	1190.625
75.0	1280.000	1289.814
90.0	1375.000	1379.088
95.0	1505.000	1432.515
99.0	1600.000	1532.734

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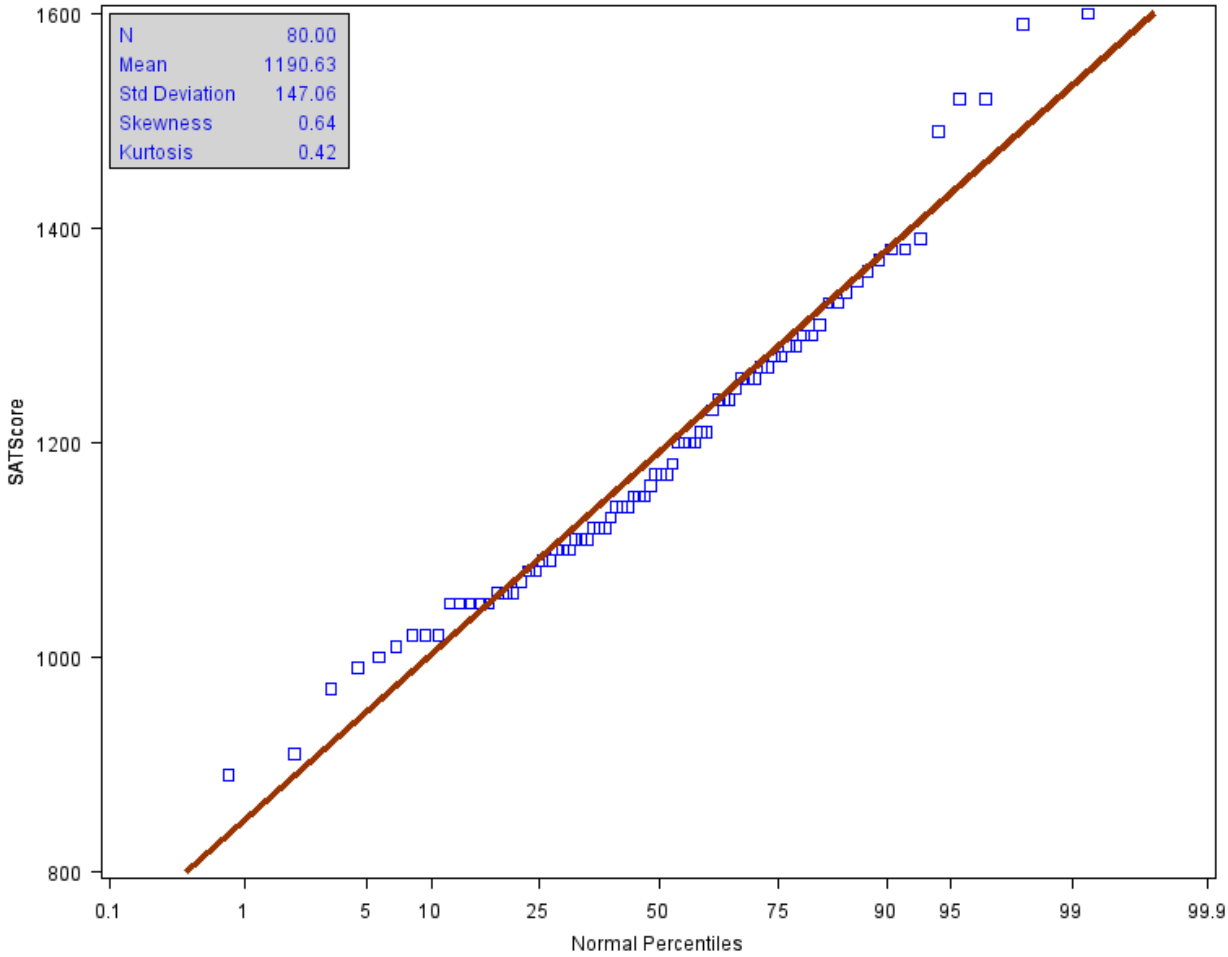

The normal probability plot is shown. The 45-degree line represents where the data values would fall if they came from a normal distribution. The circles represent the observed data values. Because the circles follow the 45-degree line in the graph, you can conclude that there does not appear to be any severe departure from normality.

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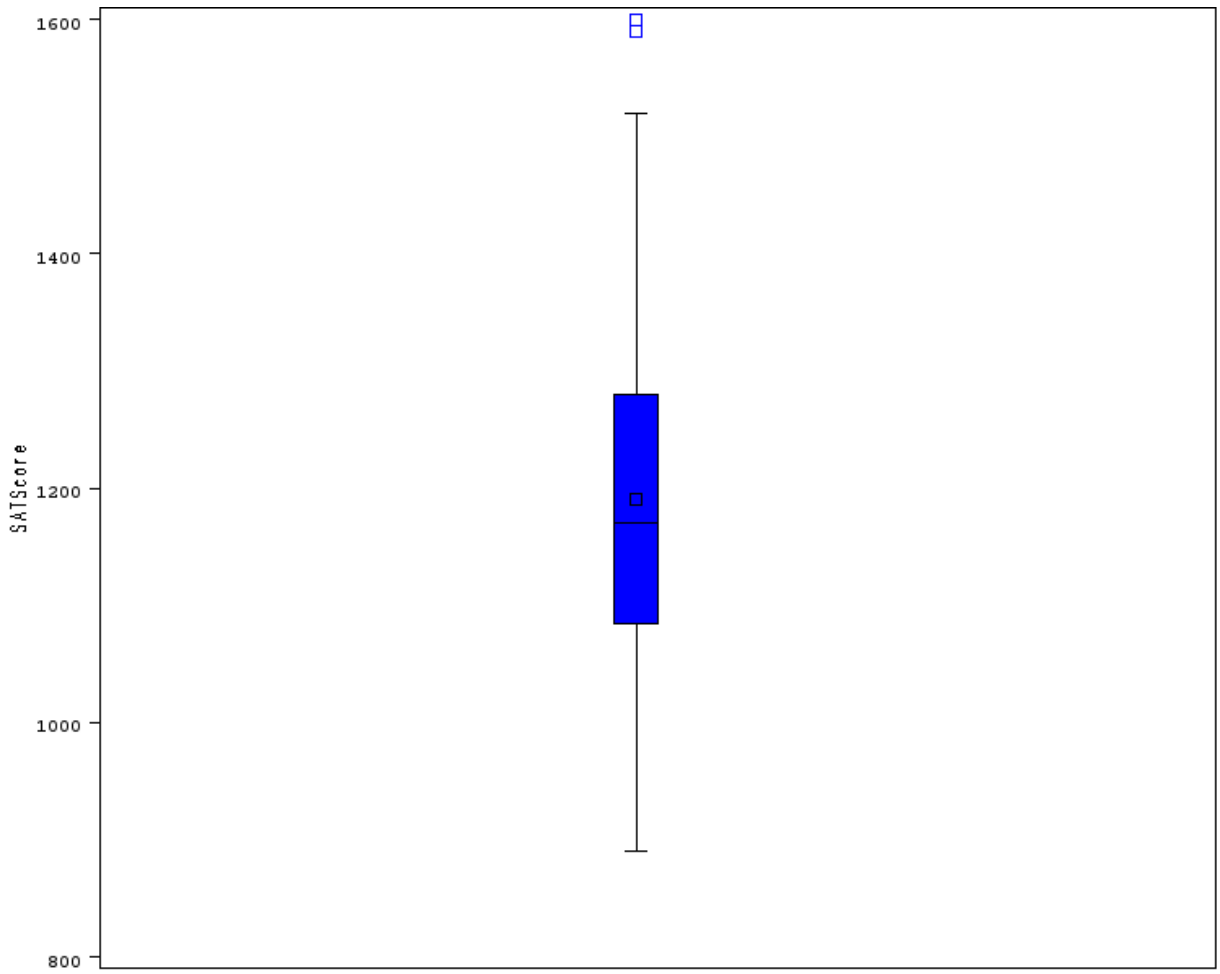

There are two outliers (values beyond 1.5 interquartile units from the box).

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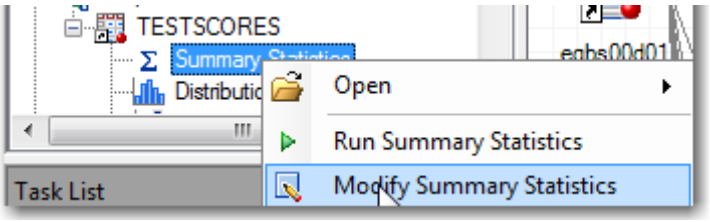
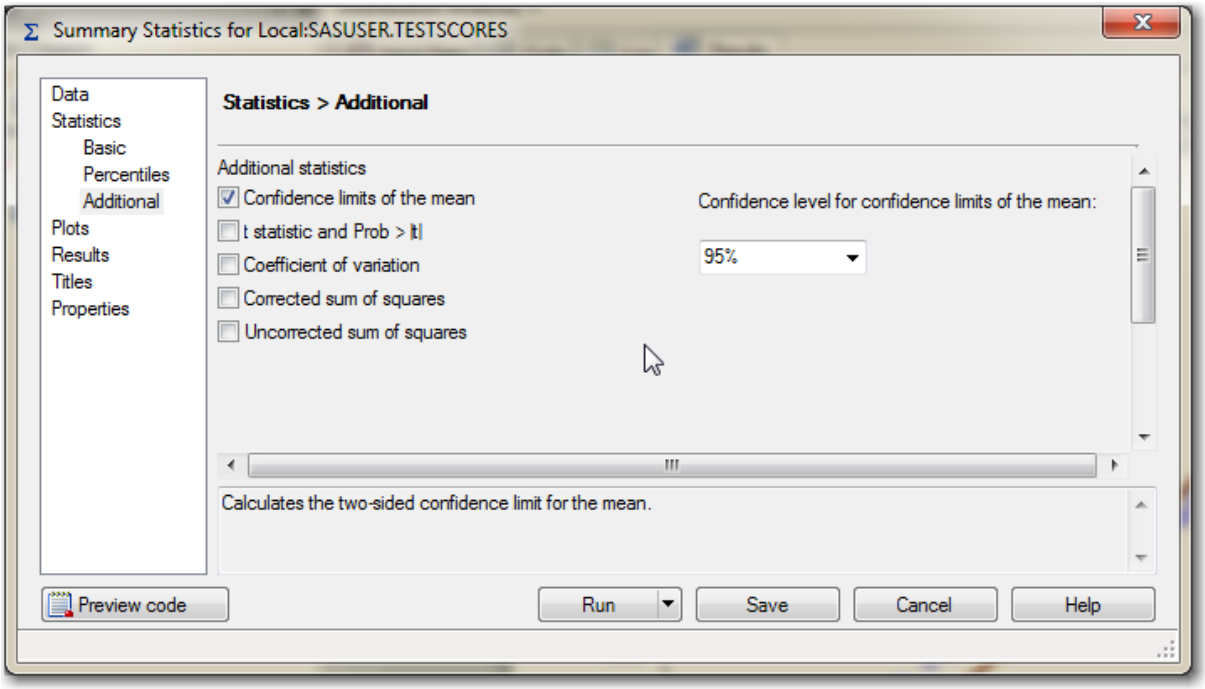
Confidence Intervals

Use the Summary Statistics task to generate a 95% confidence interval for the mean of **SATScore** in the **testscores** data set.

1. Reopen the Summary Statistics task from earlier in the section. To do so, right-click the **Summary Statistics** icon in the **Project Tree** pane and then select **Modify Summary Statistics** from the drop-down menu.



2. Select **Additional** on the left and then check **Confidence limits of the mean**. Leave the confidence level at 95%.



3. Click **Run** and then click **Yes** when asked if you want to replace the results from the previous run.

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The output is shown below.

Analysis Variable : SATScore									
Mean	Std Dev	Minimum	Maximum	N	Lower Quartile	Median	Upper Quartile	Lower 95% CL for Mean	Upper 95% CL for Mean
1190.63	147.06	890.00	1600.00	80	1085.00	1170.00	1280.00	1157.90	1223.35

In the test score example, you are 95% confident that the population mean is contained in the interval 1157.8987 and 1223.3513. Because the interval between the upper and lower limits is small from a practical point of view, you can conclude that the sample mean is a fairly precise estimate of the population mean.

How do you increase the ***precision*** of your estimate using the same confidence level? If you increase your sample size, you reduce the standard error of the sample mean and therefore reduce the width of your confidence interval. Thus, your estimate will be more precise.

Accuracy is the difference between a sample estimate and the true population value. *Precision* is the difference between a sample estimate and the mean of the estimates of all possible samples that can be taken from the population. For an unbiased estimator, precision and accuracy are the same.

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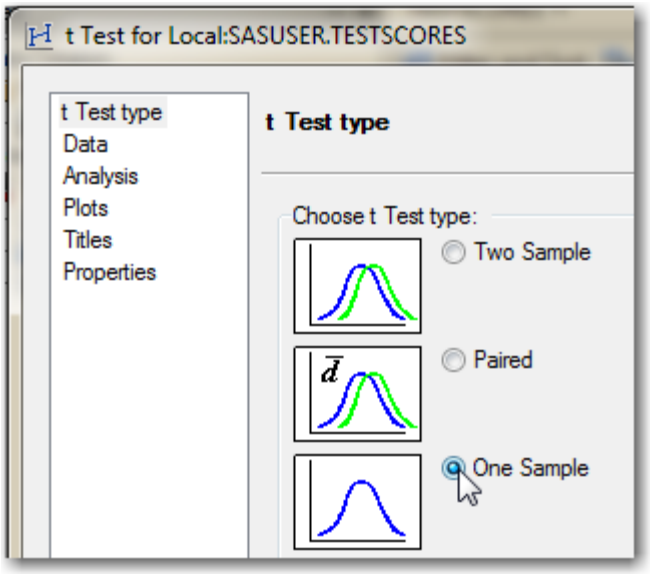
Hypothesis Testing

Example: Use the t Test task to test the hypothesis that the mean of SAT Math+Verbal score is equal to 1200.

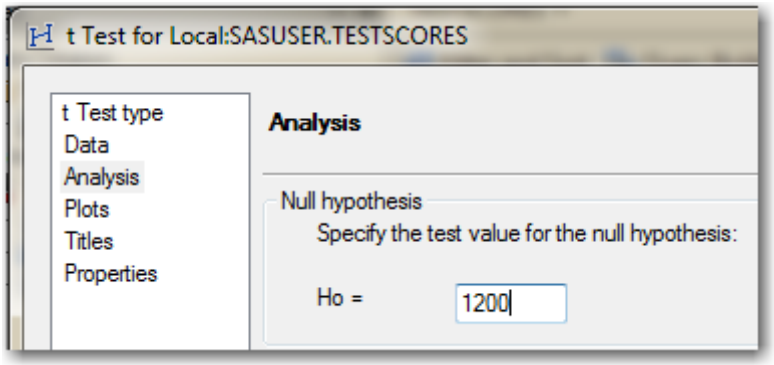
1. With the **TESTSCORES** data set selected, select **Analyze** ⇨ **ANOVA** ⇨ **t Test** from the drop-down menus.

2. Select **One Sample** from among the options.



3. After selecting **Data** on the left, assign **SATScore** to the **Analysis variables** task role.

4. With **Analysis** selected on the left, type **1200** in the box for **H0=**.



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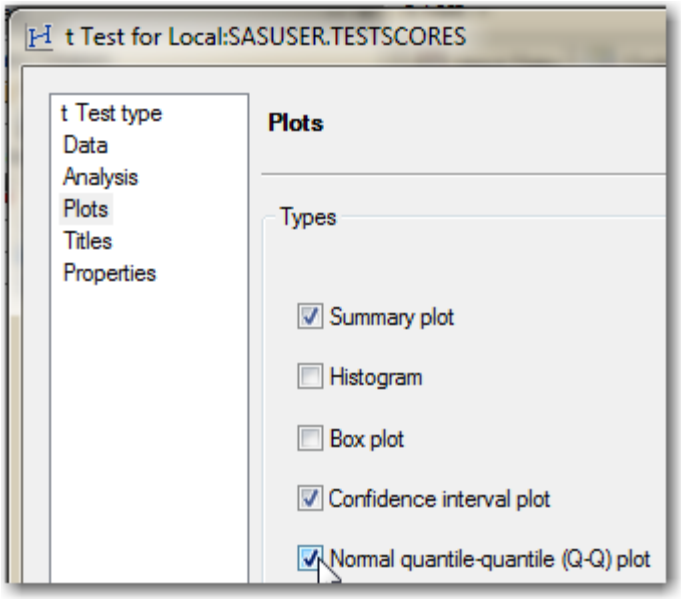
5. After selecting **Plots** on the left, check the boxes for **Summary plot**, **Confidence interval plot**, and **Normal quantile-quantile (Q-Q) plot**.



6. Click **Run**.

The t Test task Output

N	Mean	Std Dev	Std Err	Minimum	Maximum
80	1190.6	147.1	16.4416	890.0	1600.0

Summary statistics are reported. These are the same values that were obtained using both the Summary Statistics task and the Distribution Analysis task.

Mean	95% CL Mean		Std Dev	95% CL Std Dev	
1190.6	1157.9	1223.4	147.1	127.3	174.2

The confidence interval around the sample mean and sample standard deviation are reported.

DF	t Value	Pr > |t|
79	-0.57	0.5702

The *p*-value is 0.5702. As the confidence interval information suggested, this value is not statistically significant at the α=0.5 significance level.

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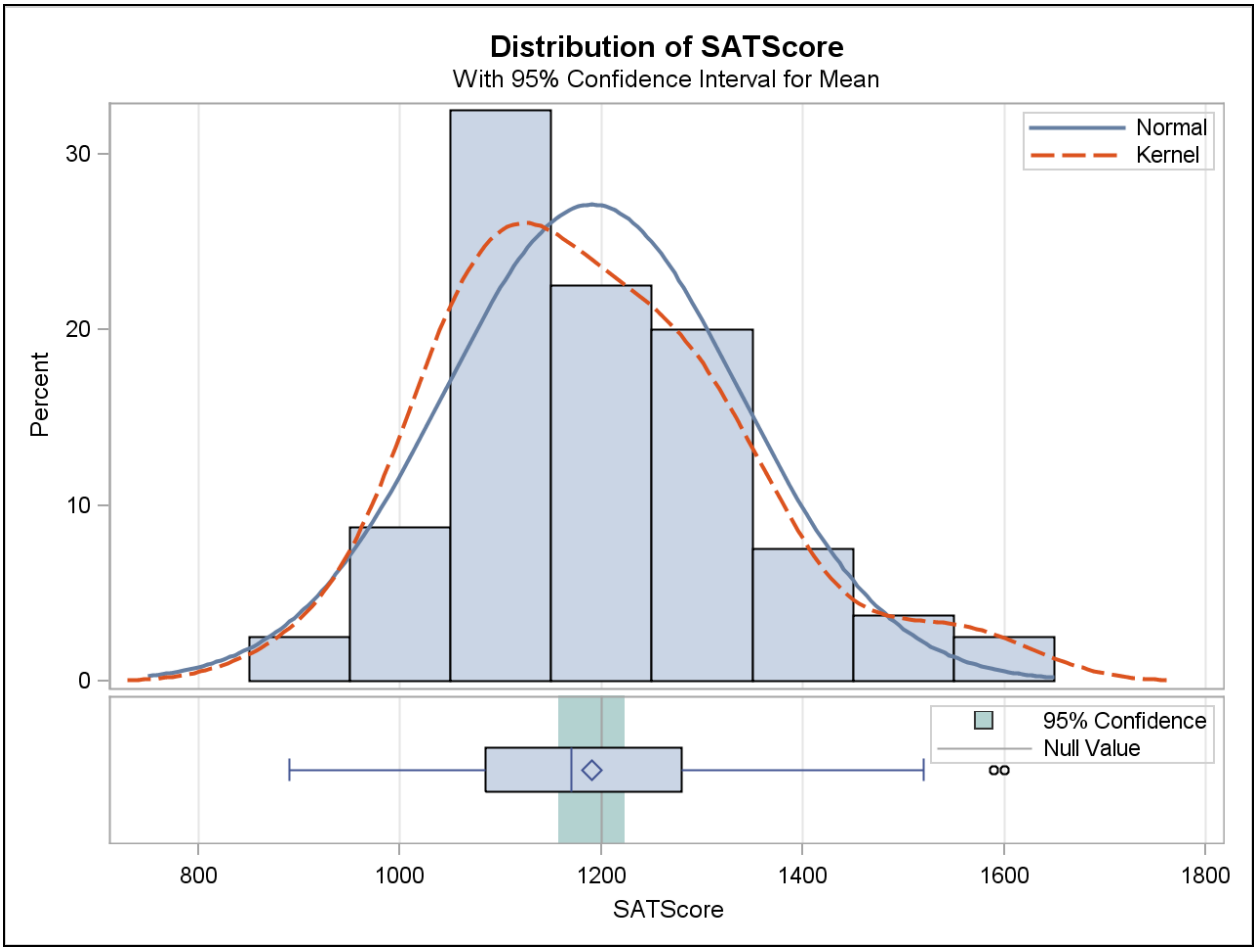
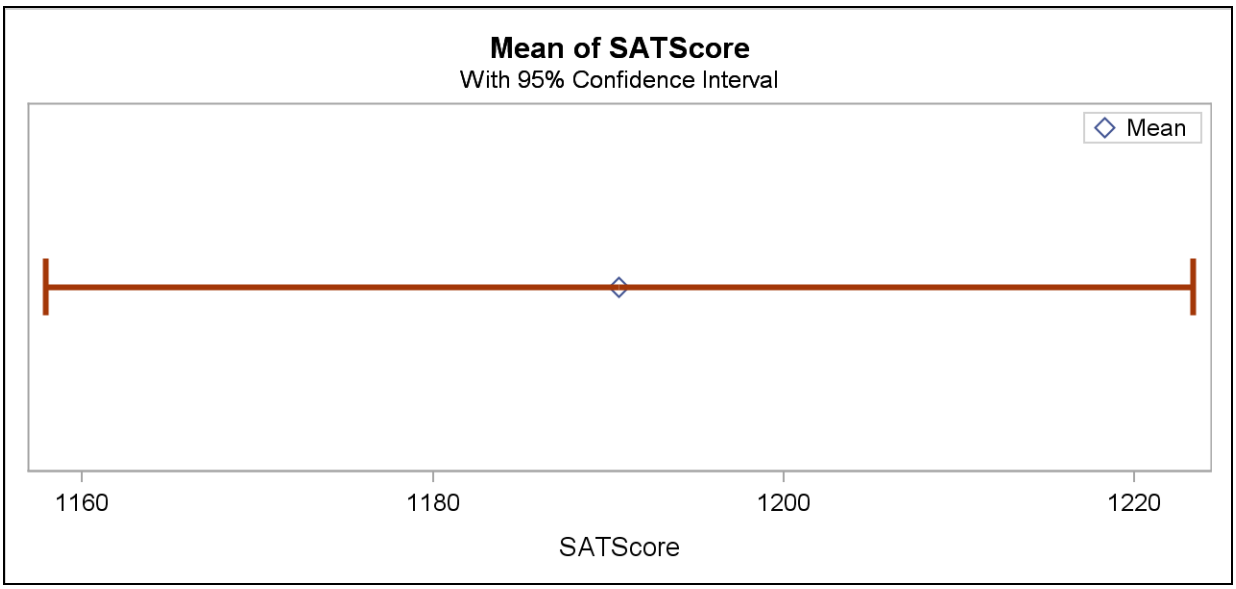


The histogram along with the normal and kernel density curves are produced on one plot, along with a horizontal box plot with a shaded confidence interval for the mean included.



The confidence interval plot gives a visual display of the confidence interval table from above.

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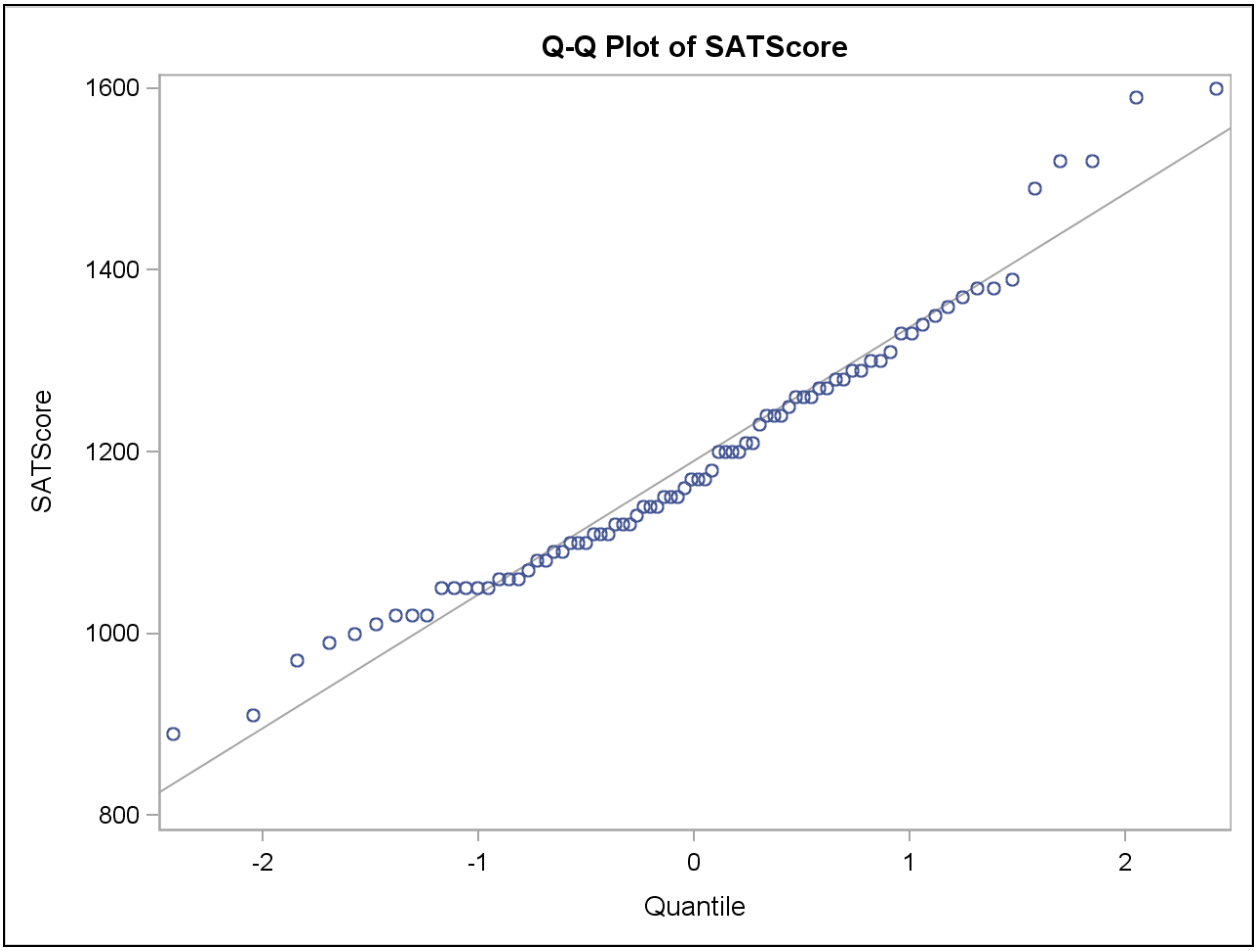

The normal quantile-quantile plot shows that the distribution of **SATScore** is approximately normal in this sample.

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